

# P-192 exact CM class-relation census

RUN-SCURVE-81c5e4 — protocol v2

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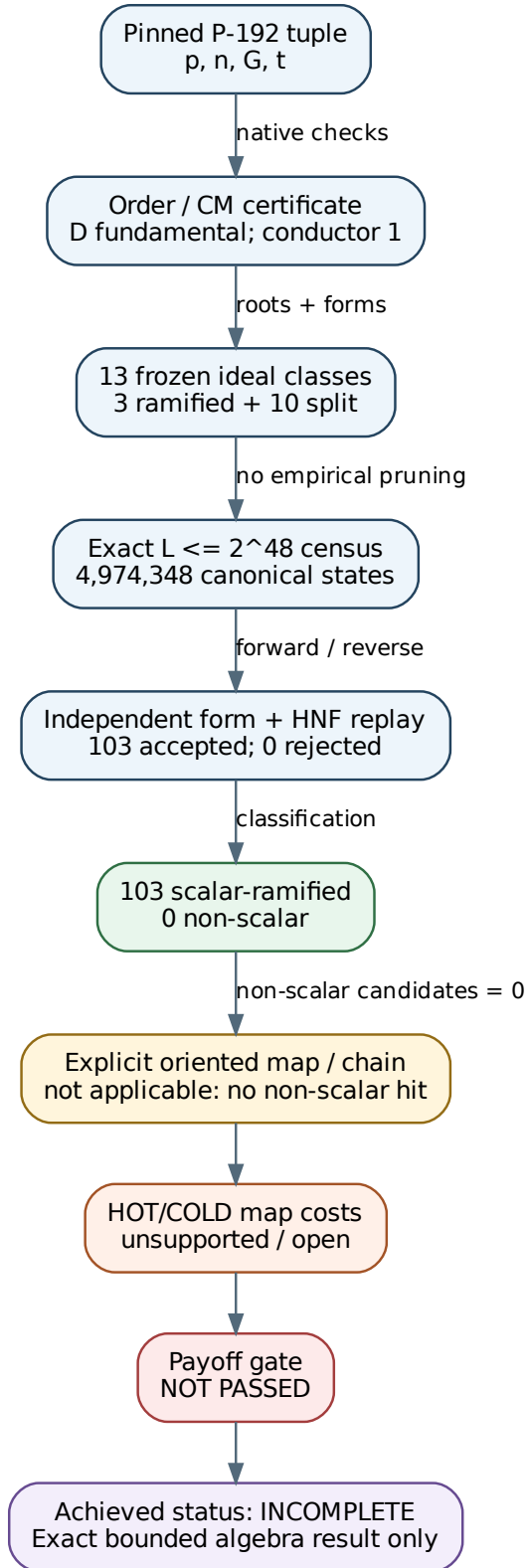
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## Outcome

**Achieved result:** complete exact algebra census; no non-scalar relation in the frozen box.

The run enumerated the preregistered P-192 class-relation search space product  $\text{ell}^{|e_{\text{ell}}|} \leq 2^{48}$  over the 13 frozen prime-ideal generators. Both enumeration orders produced the same 4,974,348 canonical states and the same set digest. Exactly 103 nonzero principal relations were found; all were scalar ramified relations, and independent replay rejected none. No non-scalar relation was found.

This is a finite algebra result, not a curve-weakness result. The full experiment remains **INCOMPLETE** because explicit globally oriented maps, point evaluation, 368 HOT/COLD calibration records, terminal isomorphisms, and a fully charged payoff comparison are unavailable. Consequently, the payoff gate did not pass and no speedup or structural P-192 weakness is asserted.



## Exact object and certificate chain

The object is the valid NIST P-192 / secp192r1 curve, not the singular companion used by the separate implementation-validation experiment. The pinned tuple, group-order checks, and CM data are recorded in order.json:

- $p$  and  $n$  pass a fixed 20-base probable-prime check and retain their SEC 2 / FIPS pinned-standard provenance; they are not presented as new recursive primality proofs.
- $G$  is on the curve,  $G \neq 0$ , and  $[n]G = 0$ ; the Hasse interval has a unique compatible multiple, fixing  $\#E(F_p)=n$ .
- $D=t^2-4p=-5*11*31*C$  is squarefree and fundamental. A recursive Pocklington certificate proves  $C$  prime, so the Frobenius conductor is one and  $\text{End}(E)=\mathbb{Z}[\pi]=\mathbb{Z}$ .
- Every non-scalar alpha in this order has a certified norm lower bound of approximately  $2^{191.941}$  (the exact integer is in order.json). This is strictly above the frozen  $2^{48}$  lane.

generators.json records all 13 oriented generators, their modular roots, reduced forms, and cross-checks. It also records passing ramified-square controls at 5, 11, and 31; a non-scalar positive control at  $D=-23$ ; and the symbolic P-192 C-degree control. The symbolic control is outside the buildable factor base and does not imply an attack.

## Exact census and replay

The authoritative search certificate is relations.json, and the independent verifier result is verification.json. The extracted boundary view is exact-boundary-manifest.json.

| quantity                               | forward   | reverse   | expected  |
|----------------------------------------|-----------|-----------|-----------|
| oriented vectors                       | 9,948,061 | 9,948,061 | 9,948,061 |
| conjugation-fixed vectors              | 635       | 635       | 635       |
| canonical vectors including zero       | 4,974,348 | 4,974,348 | 4,974,348 |
| canonical nonzero vectors              | 4,974,347 | 4,974,347 | 4,974,347 |
| scalar-principal states including zero | 104       | 104       | 104       |
| exact $L=2^{48}$ vectors               | 0         | 0         | 0         |

The forward and reverse canonical-set digest is identical (xor begins fa31a647 and ends 4573e; sum begins 8eedade9 and ends 51806). The complete values are recorded in exact-boundary-manifest.json.

The exact-boundary encoding is empty because every generator prime is odd; its SHA-256 is the standard empty-input digest (e3b0c442...b855; complete value in the boundary manifest). The radius-16 control examined the complete 2,997,268-state radius-8 half ball, found zero duplicate classes after the specified scalar-ramified quotient, and found no extra relation.

For each of the 103 retained relations, the verifier recomputed the exponent vector, exact degree, reduced form,  $\alpha=(u,v)$ , norm, scalar action, ideal HNF, and principal HNF. It accepted all 103 and rejected none. The zero vector was excluded separately. The result therefore says exactly that the frozen box contains no non-scalar class relation; it does not cover larger degree, other factor bases, other attack families, or all curves in the isogeny class.

## Payoff and transfer obligations

An abstract form identity does not transfer a discrete-log problem for free. The required chain is: a globally Frobenius-oriented ideal, an explicit kernel/map, subgroup preservation, evaluated points, a terminal isomorphism, independent replay, and a costed advantage over a declared baseline. Here the search produced no non-scalar candidate on which to attempt that chain.

weights.json truthfully records unsupported\_open: zero of the required 368 calibration records were emitted because explicit P-192 maps are not implemented. operation-counts.json records the HOT

and COLD tuples as null rather than fabricating zeros. No wall-clock time or degree-linear proxy is substituted for those missing operation costs.

Thus the certificate field `claim_ceiling=NO_WEAKNESS_FOUND_WITHIN_SCOPE` is only the maximum permitted claim. The achieved full-experiment status is `incomplete-map-calibration-and-payoff-open`.

## Execution history and defects found

The first launcher attempt failed before child execution because `/usr/bin/time` was absent; it consumed no experimental run. Two native execution attempts then found real harness defects and are preserved:

1. Attempt 1 exposed an incorrect Bézout coefficient and a duplicated factor in generic binary-quadratic-form composition. The first corrupted product was the positive 103-by-107 class; the next 113 composition triggered the invariant assertion. The correction added exact divisibility checks and full-census regressions.
2. Attempt 2 completed the census but the exact verifier rejected six `log2_degree` values after JSON parsing changed them by one ULP. Enabling `serde_json's float_roundtrip` parser retained exact equality. Regressions now preserve the formerly pathological value and reject one-ULP and NaN mutations.
3. Attempt 3, at code commit `78ba2da...e6669`, completed all five frozen phases with exit status zero. The release binary SHA-256 is `91a02361...c0111c`; `run.yaml` and the manifest retain the complete values.

Preserving these failures is part of the evidence: neither a panic nor a serialization mismatch was reclassified as a mathematical result.

## Resources and statistical confidence

`resource-metrics.json` derives directly from the five records in `isolation.jsonl`. All phases were pinned to CPUs 1-4 and marked uncontended. Their aggregate wall time was 39.733819654 seconds, with a maximum phase RSS of 639,024 KiB. The exact search used 28.239448942 seconds; full certificate replay used 11.462448644 seconds.

These are single-run engineering observations. They have no sampling error bar or timing confidence interval and support no throughput comparison. The algebraic completeness claim instead obtains confidence from exhaustive coverage under a frozen boundary, two enumeration orders with matching digests, exact independent HNF/form replay, mutation regressions, theorem and positive controls, and a SHA-256 artifact manifest.

## Verdict

Supported:

- complete exact algebra census; no non-scalar relation in the frozen box;
- 103 nonzero scalar-ramified relations, all independently replayed;
- no evidence of a useful class relation within the frozen factor base and norm boundary.

Not supported:

- that P-192 is structurally weak or its discrete logarithm is easier;
- that no useful relation exists outside this bounded search;
- that a relation on another curve automatically transfers an attack;
- any HOT/COLD map-cost, timing-speedup, or production-key claim.

The strongest achieved verdict is therefore **INCOMPLETE exact-algebra pass**, with no non-scalar relation in scope—not a declaration that P-192 is weak and not a global proof of strength.